

Learning with Hidden Variables

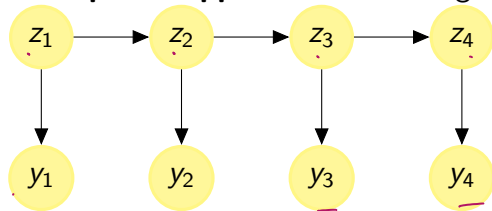
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Learning with hidden variables

Suppose only a subset of variables are observed. Other variables are hidden variables. How to learn the parameters of the graphical model?

Example of application: Training HMM



Here, some variables $z_1^i, z_2^i, \dots, z_m^i$ are not present in D but is present in the graphical model.

Learning parameters with hidden variables

Let θ be the parameters of the graphical model.

$$P_{\theta, G}(y_1, y_2, \dots, y_n, z_1, z_2, \dots, z_m | \mathbf{x})$$

Suppose $D = \{(\mathbf{x}^i, \mathbf{y}^i) : i = 1, \dots, N\}$ is our dataset
Our goal during training is to find,

$$\begin{aligned} \theta^{ML} &= \arg \max_{\theta} \sum_{i=1}^N \log P_{\theta}(\mathbf{y}^i | \mathbf{x}^i) \\ &= \arg \max_{\theta} \sum_{i=1}^N \log \sum_{\mathbf{z}} P_{\theta, G}(\mathbf{y}^i, \mathbf{z} | \mathbf{x}^i) \end{aligned}$$

The summation over $\mathbf{z}$ within the log make optimization difficult.
Hence we approximate this objective. We will apply ideas from variational approximation to solve this problem. We will see that it will give rise to the well-known EM algorithm.

Variational Approach

We rewrite the original optimization in terms of new auxiliary variables that we introduce.

$$\begin{aligned} & \max_{\theta} \sum_{i=1}^N \log \sum_{\mathbf{z}: \mathbf{z}_1, \dots, \mathbf{z}_m} P(\mathbf{y}^i, \mathbf{z} | \theta, \mathbf{x}^i) \\ & \equiv \max_{\theta} \sum_{i=1}^N \max_{q_{i,z}: \sum_z q_{i,z} = 1} \sum_z q_{i,z} \log P(\mathbf{y}^i, \mathbf{z} | \theta, \mathbf{x}^i) - \sum_z q_{i,z} \log q_{i,z} \end{aligned}$$

The advantage of this rewriting is that now we do not have summation within the log.

We have two maximization problems to solve: over θ and over q variables.

The inner one can be solved in closed form for fixed value of θ .

The outer one can be solved like normal MLL training without hidden variables.

Variational approach (Proof)

We will show that:

$$\log \sum_{z=1}^k g(y, z) \equiv \max_{q_1, q_2, \dots, q_k} \sum_{z=1}^k q_z \log g(y, z) - \sum_z q_z \log q_z$$

s.t. $\sum_{z=1}^k q_z = 1$ and $q_z \geq 0$

Handwritten notes:
- A checkmark is above the maximization symbol.
- The expression $q_z^* = \frac{g(y, z)}{\sum_j g(y, j)}$ is written on the left.
- The constraint is enclosed in a large pink bracket.

where $q_1, \dots, q_k$ are auxiliary variables.

Define:

$$Q(\underline{q}, g) = \sum_{z=1}^k q_z \log g(y, z) - \sum_z q_z \log q_z$$

Handwritten notes:
- The vector $\underline{q}$ is underlined in pink.
- The entire expression is enclosed in a large pink oval.

Variational approximation(Continued)

Let $L(Q, \lambda) = Q(q, g) + \lambda(\sum_z q_z - 1)$, then

$$\begin{aligned} \rightarrow \frac{\partial L(Q, \lambda)}{\partial q_j} &= \log g(y, j) - \log q_j - 1 + \lambda \\ &\Rightarrow q_j^* \equiv \frac{g(y, j)}{\sum_z g(y, z)} \end{aligned}$$

Substituting q^* in the original maximization equation, we get $\log \sum_{z=1}^k g(y, z)$.

Hence LHS=RHS.

Applying variational approximation

$$\begin{aligned}\theta^{ML} &\equiv \arg \max_{\theta} \sum_{i=1}^N \log \sum_{\mathbf{z}: \mathbf{z}_1, \dots, \mathbf{z}_m} P(\mathbf{y}^i | \theta, \mathbf{x}^i) \\ &\equiv \arg \max_{\theta} \sum_{i=1}^N \max_{q_{i,z}: \sum_z q_{i,z}=1} \sum_{\mathbf{z}} q_{i,z} \log P(\mathbf{y}^i, \mathbf{z} | \theta, \mathbf{x}^i) - \sum_{\mathbf{z}} q_{i,z} \log q_{i,z}\end{aligned}$$

A change of notation

$x \rightarrow \phi$
 $y \rightarrow x$

Without loss of generality assume generative models, that is $\mathbf{x}$ is absent. Also, to align with existing literature $\mathbf{y}$ is expressed as $\mathbf{x}$.

$$\begin{aligned}
 \underline{\theta^{ML}} &\equiv \arg \max_{\theta} \sum_{i=1}^N \log \sum_{\mathbf{z}: z_1, \dots, z_m} P_{\theta}(\mathbf{x}^i) \\
 &\equiv \arg \max_{\theta} \sum_{i=1}^N \max_{q_{i,z}: \sum_z q_{i,z}=1} \sum_z q_{i,z} \log P_{\theta}(\mathbf{x}^i, \mathbf{z}) - \sum_z q_{i,z} \log q_{i,z} \\
 &\equiv \arg \max_{\theta} \sum_{i=1}^N \max_{q_{i,z}: \sum_z q_{i,z}=1} \sum_z q_{i,z} \log P_{\theta}(\mathbf{x}^i | \mathbf{z}) + \sum_z q_{i,z} \log \frac{P_{\theta}(\mathbf{z})}{q_{i,z}}
 \end{aligned}$$

Handwritten notes: $P_{\theta}(\mathbf{x}^i | \mathbf{z}) P_{\theta}(\mathbf{z})$ (circled around the second term), $P_{\theta}(\mathbf{z})$ (circled around the third term).

How to solve this joint optimization problem?

Variational Approximation (Alternative Rewrites)

Since $q_{i,z} \geq 0$, $\sum_z q_{i,z} = 1$, we will view $q_{i,z}$ as distribution $\underline{q}(\underline{z}|\underline{x}^i)$. Works for both discrete and continuous $\underline{z}$. For continuous $\underline{z}$ we get.

$$\begin{aligned}
 & \max_{\theta} \sum_{i=1}^N \log P_{\theta}(\mathbf{x}^i) \\
 &= \max_{\theta} \sum_{i=1}^N \max_{q(\mathbf{z}|\mathbf{x}^i)} \int_{\mathbf{z}} (q(\mathbf{z}|\mathbf{x}^i) \log P_{\theta}(\mathbf{x}^i|\mathbf{z}) P_{\theta}(\mathbf{z}) - q(\mathbf{z}|\mathbf{x}^i) \log q(\mathbf{z}|\mathbf{x}^i)) d\mathbf{z} \\
 &= \max_{\theta} \sum_{i=1}^N \max_{q(\mathbf{z}|\mathbf{x}^i)} \underbrace{\int_{\mathbf{z}} q(\mathbf{z}|\mathbf{x}^i) \log P_{\theta}(\mathbf{x}^i|\mathbf{z}) d\mathbf{z}}_{\text{Expected Likelihood}} - \underbrace{\int_{\mathbf{z}} q(\mathbf{z}|\mathbf{x}^i) \log \frac{q(\mathbf{z}|\mathbf{x}^i)}{P_{\theta}(\mathbf{z})} d\mathbf{z}}_{\text{KL-distance}(q(\mathbf{z}|\mathbf{x}^i), P_{\theta}(\mathbf{z}))} \\
 &= \max_{\theta} \sum_{i=1}^N \max_{q(\mathbf{z}|\mathbf{x}^i)} \mathbb{E}_{q(\mathbf{z}|\mathbf{x}^i)} \log P_{\theta}(\mathbf{x}^i|\mathbf{z}) - \text{KL-distance}(q(\mathbf{z}|\mathbf{x}^i), P_{\theta}(\mathbf{z}))
 \end{aligned}$$

Methods of solving for $q(\mathbf{z}|\mathbf{x}^i)$

- 1 Expectation Maximization (EM) Algorithm: Alternately solve for θ and $q(\mathbf{z}|\mathbf{x}^i)$ while keeping the other fixed.
Not tractable when $\mathbf{z}$ is high-dimensional and continuous.
- 2 Variational auto-encoders (VAE): Approximate $q(\mathbf{z}|\mathbf{x}^i)$ as output of another neural network that takes as input $(\mathbf{x}^i)$ and a distribution $q_\phi(\mathbf{z}|\mathbf{x}^i)$
- 3 Denoising Diffusion models: Break up the $P_\theta(\mathbf{x}^i|\mathbf{z})$ term into multiple smaller steps, so that for each step a simple fixed $q(\mathbf{z}|\mathbf{x}^i)$ can be assumed.
- 4 Normalizing Flows (NF): Assume a special invertible function to implement $P_\theta(\mathbf{x}^i|\mathbf{z})$ so that optimal $q(\mathbf{z}|\mathbf{x}^i)$ can be computed in closed form.

$t=1$: arbitrary initial parameter θ^1

EM algorithm

for $t=1$ to T .

E-Step: Solve for $q(\mathbf{z}|\mathbf{x}^i)$ keeping θ fixed at θ^t

$$\begin{aligned} q^t(\mathbf{z}|\mathbf{x}^i) &\equiv \frac{P(\mathbf{x}^i, \mathbf{z}|\theta^t)}{\sum_{\mathbf{z}} P(\mathbf{x}^i, \mathbf{z}|\theta^t)} \\ &\equiv P(\mathbf{z}|\theta^t, \mathbf{x}^i) \end{aligned}$$

Hence $q^t(\mathbf{z}|\mathbf{x}^i)$ is the posterior distribution of hidden variable at time t .

M-Step: Solve for θ , keeping $q(\mathbf{z}|\mathbf{x}^i)$ fixed to $q^t(\mathbf{z}|\mathbf{x}^i)$ The problem becomes

$$\theta^{t+1} = \arg \max_{\theta} \sum_i \sum_{\mathbf{z}} q^t(\mathbf{z}|\mathbf{x}^i) \log P(\mathbf{x}^i, \mathbf{z}|\theta)$$

which is concave in θ and can be often solved in closed form. (Example: HMM)

For graphical models, the $q(\mathbf{z}|\mathbf{x}^i)$ variables can be factorized over cliques

Example to understand notation

$$\underline{\mathbf{z}} = [\underline{z}_1, \dots, \underline{z}_4]$$

Assume each $\underline{z}_i, \underline{x}_i$ is binary for all i .

Assume $\underline{\mathbf{x}}^i = [0, 0, 0, 1]$

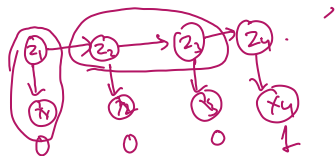
For each i, t , we will have $\underline{2}^4$ q variables corresponding to each assignment of values to $\underline{\mathbf{z}}$

$$q_{i,0000}^t, q_{i,0001}^t, \dots, q_{i,1111}^t$$

Example:

$$q_{i,0000}^t = \frac{P(\underline{\mathbf{x}}^i, \underline{\mathbf{z}}=[0000]|\theta^t)}{P(\underline{\mathbf{x}}^i, \underline{\mathbf{z}}=[0000]|\theta^t) + P(\underline{\mathbf{x}}^i, \underline{\mathbf{z}}=[0001]|\theta^t) + \dots + P(\underline{\mathbf{x}}^i, \underline{\mathbf{z}}=[1111]|\theta^t)}$$

$$= P(\underline{\mathbf{z}}=[0,0,0,0] \mid \underline{\mathbf{x}}=[0,0,0,1])$$



$$\theta^t = \begin{cases} P(z_t | z_{t-1}) \\ P(x_t | z_t) \\ P(z_1) \end{cases}$$

Take Aways for Learning with Hidden variables

- ① When not all variables are observed in each training sample, learning of potentials expressing their interaction becomes complicated.
- ② EM algorithm is an efficient option: it alternates between estimating probability distribution over hidden variables and solving for potentials with weighted instances.
- ③ Variational approximation provides a formal justification for the EM algorithm and is more broadly applicable as we will see next.
- ④ For graphical models: the distribution over hidden variables can be factorized.

Deep Learning Models for High-dimensional generation

Task: Train an ML model to generate high-dimensional objects given a context $\mathbf{c}$.

Assume $\mathcal{X} \in \mathbb{R}^n$ denotes the space of objects where n is large.

Example:

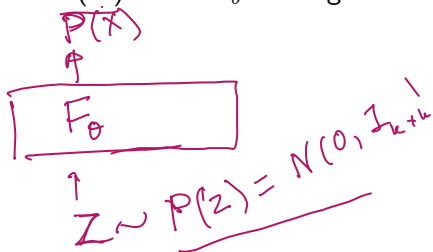
- 1 Images, where $n = 1024 \times 1024 \times 3$ to denote with 1024 pixels on each side and 3 channels (red, green, blue). The context could be a piece of text describing the image.
- 2 Audio waveform, Videos..

Several impressive examples

- 1 Imagen: <https://imagen.research.google/>
- 2 Dall-E: <https://labs.openai.com/>
- 3 Stable Diffusion: <https://stablediffusionweb.com/>

Problem Statement

- Given: A dataset $D = \{(\underline{\mathbf{c}}^1, \underline{\mathbf{x}}^1) \dots, (\underline{\mathbf{c}}^N, \underline{\mathbf{x}}^N)\}$ of N objects assumed to be samples from some unknown distribution of objects $P(\mathcal{C}, \mathcal{X})$.
- Latent Variable Models
 - Assume a latent variable space $\mathcal{Z} \in \mathbb{R}^k$ with a known simple distribution $P(\mathcal{Z})$ e.g. standard Gaussian from which it is easy to sample.
 - A parametric network $F_\theta : \mathcal{Z}, \mathcal{C} \mapsto P(\mathcal{X})$ where F_θ is a high capacity deep neural network.



Problem statement

- Training
 - ① Learn the parameters θ using data D .
 - ② Several objectives based on the type of the model.
- Generation/Deployment: Given a $\mathbf{c}$
 - ① Sample a $\mathbf{z} \sim P(\mathcal{Z})$
 - ② Return $\mathbf{x} \sim F_{\theta}(\mathbf{z}, \mathbf{c})$

Some Assumptions

- ① We will initially consider continuous variables. Generation of discrete objects, example language, will be discussed a bit later.
- ② We drop the context (c) and focus on unconditional generation. Context can be included easily later.

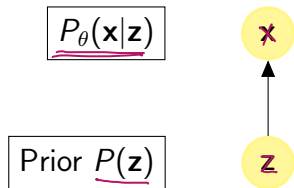
Types of Latent Variable Generative Models

Different ways of training F_θ

- 1 Variational Auto Encoders(VAEs) —
- 2 Normalizing Flows (NF) ~
- 3 Generative Adversarial Networks(GANs) ✕
- 4 Denoising Diffusion models →

Variational Auto Encoders(VAEs)

A probabilistic approach for training F_{θ} which is assumed to model $P_{\theta}(\mathbf{x}|\mathbf{z})$. Thus, the underlying graphical model is:



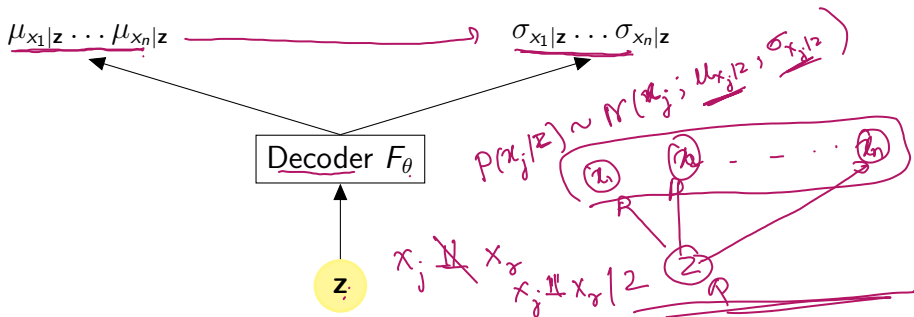
During training we learn θ so as to maximize the likelihood of the training data.

Representing $P_{\theta}(\mathbf{x}|\mathbf{z})$

The $P(\mathbf{x}|\mathbf{z})$ distribution is complicated (it has to generate complex objects, example images, time series, etc).

We choose to make it a conditional Gaussian distribution whose parameters are obtained from a deep network.

$$P(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; \mu = [\mu_{x_1|\mathbf{z}} \dots \mu_{x_n|\mathbf{z}}], \Sigma = \text{diagonal}[\sigma_{x_1|\mathbf{z}} \dots \sigma_{x_n|\mathbf{z}}])$$



Training VAEs

Training data $D = \{\mathbf{x}^i\}_{i=1}^N$ is observed and $\mathbf{z}$ is hidden. Marginalize over $\mathbf{z}$ to compute $P(\mathbf{x})$.

$$\max_{\theta} \sum_{i=1}^N \log P_{\theta}(\mathbf{x}^i) = \max_{\theta} \sum_{i=1}^N \log \int_{\mathbf{z}} P_{\theta}(\mathbf{x}|\mathbf{z}) P_{\theta}(\mathbf{z}) d\mathbf{z}.$$

$\mathbf{z} \in R^k, \mathbf{x} \in R^n$ Integrating over all $\mathbf{z}$ is intractable. How to tackle this?

Use variational approximation of the objective.

Variational approximation

$$\begin{aligned}
 \sum_{i=1}^N \log P_{\theta}(\mathbf{x}^i) &= \sum_{i=1}^N \log \int_{\mathbf{z}} P_{\theta}(\mathbf{x}^i | \mathbf{z}) P_{\theta}(\mathbf{z}) d\mathbf{z} \\
 &= \sum_{i=1}^N \max_{q_{i,z}} \int_{\mathbf{z}} (q_{i,z} \log P_{\theta}(\mathbf{x}^i | \mathbf{z}) P_{\theta}(\mathbf{z}) - q_{i,z} \log q_{i,z}) d\mathbf{z} \\
 \text{s.t. } q_{i,z} &\geq 0, \int q_{i,z} = 1
 \end{aligned}$$

EM

$$q(\mathbf{z} | \mathbf{x}^i) = P(\mathbf{z}) P_{\theta}(\mathbf{x}^i | \mathbf{z})$$

$$\int P(\mathbf{z}) P_{\theta}(\mathbf{x}^i | \mathbf{z}) d\mathbf{z}$$

intractable

Cannot afford to solve the inner maximization for each $i, \mathbf{z}$! How does the variational rewrite help?